

11. Rating Political Candidates A political scientist is studying the relationship between the voter image of a conservative political candidate and the distance (in kilometers) between the residences of the voter and the candidate. Each of 12 voters rated the candidate on a scale of 1–20.

Voter	Rating	Distance	$d = (R - 7)$	d^2
1	12 ^{7.5}	75 ³	4.5	20.25
2	7 ⁴	165 ⁷	-3	9
3	5 ³	300 ¹²	-9	81
4	19 ¹²	15 ¹	11	121
5	17 ¹⁰	180 ⁸	2	4
6	12 ^{7.5}	240 ¹¹	-3.5	12.25
7	9 ⁶	120 ⁴	2	4
8	18 ¹¹	60 ²	9	81
9	3 ¹	230 ¹⁰	-9	81
10	8 ⁵	200 ⁹	4	16
11	15 ⁹	130 ^{5.5}	3.5	12.25
12	4 ²	130 ^{5.5}	-3.5	12.25
			$\sum d^2 = 454$	

- Calculate Spearman's rank correlation coefficient r_s .
- Do these data provide sufficient evidence to indicate a negative rank correlation between rating and distance?

$$1a) r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} = 1 - \frac{6(454)}{12(144 - 1)} = 1 - \frac{2724}{12 \times 143} = -0.5874$$

$$1b) H_0: \rho_s \geq 0$$

$$H_a: \rho_s < 0$$

when $n=12$, $\alpha=0.05$, 临界值 = 0.497

$$RR: \{R \leq R_\alpha = -0.497\}$$

$$\because -0.587 < -0.497, \in RR$$

→ Reject H_0 , have sufficient evidence to indicate.

12. Competitive Running Is the number of years of competitive running experience related to a runner's distance running performance? The data on nine runners, obtained from a study by Scott Powers and colleagues, are shown in the table:⁸

Runner	Years of Competitive Running	10-Kilometer Finish Time (minutes)
1	9	33.15
2	13	33.33
3	5	33.50
4	7	33.55
5	12	33.73
6	6	33.86
7	4	33.90
8	5	34.15
9	3	34.90

 $d = x - y$ d^2

6

36

7

49

0.5

0.25

2

4

3

9

-1

1

-5

25

-4.5

20.25

-8.0

64

 $\sum d^2 = 208.5$

- a. Calculate the rank correlation coefficient between years of competitive running and a runner's finish time in the 10-kilometer race.
- b. Do the data provide evidence to indicate a significant rank correlation between the two variables? Test using $\alpha = .05$.

$$(a) \quad r_s = 1 - \frac{6(208.5)}{9(9^2 - 1)} = 1 - \frac{1251}{9 \times 80} = -0.7375$$

$$(b) \quad H_0: \rho_s = 0 \\ H_a: \rho_s \neq 0$$

$$n = 9, \alpha = 0.05 \text{ at}$$

$$r_{\text{crit}} = 0.683$$

$$\rightarrow \text{when } |r_s| \geq 0.683, \in \text{RR}$$

$$\because 0.7375 > 0.683, \in \text{RR}$$

$$\rightarrow \text{Reject } H_0, \text{ have sufficient evidence.}$$